

MANOJ DATTATREYA MYNENI

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PROFESSIONAL EXPERIENCE

Software Engineer

Jun 2025 – Present

University of Illinois Chicago - Bits Network Lab

Chicago, IL

- Worked on Convex, a decentralized execution engine implementing a “Stateful Internet,” featuring a global-state VM, cryptographically secured digital assets, smart contracts, and a high-throughput Convergent Proof-of-Stake consensus protocol. — **GitHub Repository**
- Contributed as a backend engineer on the Convex Virtual Machine and peer-to-peer network stack, working on CVM execution, data structures, Etch database operations, peer synchronization, and CLI tooling.
- Improved CVM execution stability, optimized peer sync latency, enhanced transaction-processing efficiency, and contributed code that strengthened the reliability and performance of the distributed Convex test network.
- **Tech Stack:** Java, JVM, Convex Virtual Machine (CVM), Convergent Proof-of-Stake (CPoS), Etch Database, NIO Networking, gRPC/REST APIs, CLI Tools, Distributed Consensus, Git
- CloudSweeper is a security tool that helps users analyze and manage cloud-storage risks by combining file-explorer views, metadata inspection, machine-learning classification (Aletheia), and treemap visualizations to identify sensitive or expendable files across Google Drive/Dropbox.— **Website**
- Worked on the backend components supporting metadata extraction, file-risk analysis, visualization pipelines, and local inference logic. Integrated the File Explorer, DataGrid, and WebTreeMap modules into a unified workflow for cloud-file inspection.
- Built core C++ modules for file parsing, metadata handling, and structured categorization. Added debugging tools and leak-free memory management, improving stability, classification accuracy, and overall runtime reliability of the Cloudsweeper system.
- **Tech Stack:** C++, Qt Framework, File-System APIs, JSON/CSV Parsers, Python, DataGrid and WebTreeMap UI modules, Aletheia ML Classifier, Valgrind (Memory Debugging), Git, Linux

Robotics Software Engineer

Jan 2025 – Present

University of Illinois Chicago - Learning + Interest + Technology (LIT) Lab

Remote

- Developed and optimized **C++ modules** in **ROS navigation**, **perception**, and **localization** for autonomous robots(**Starship**), integrated **motion planning** and **multi-sensor fusion**, and improved obstacle detection, path accuracy, and overall motion control reliability.
- Spearheaded calibration and tuning of **LIDAR**, **GPS**, and **ultrasonic sensors** during an autonomous robot navigation project, refined sensor fusion filters, and conducted **A/B testing** to improve localization precision and validated a **20%** increase in path-planning accuracy.
- Diagnosed subsystem issues in autonomous robot operations by analyzing **telemetry logs**, evaluated **CAN bus** communication, and applied **ROS debugging tools**. Reduced critical failure incidents by **28%** and improved fleet stability across operational deployments.
- **Collaborated** with hardware researchers on electrical, mechanical, and firmware integration, executed **software updates**, safety validations, and **actuator control optimizations** to ensure seamless interaction between hardware and software components.

AI Software Engineer

Jun 2024 – Jan 2025

PeakMind LLC (Mental Health Application) — Google-Funded Startup

Chicago, IL

- Built core features for a mental-health application offering AI-guided sessions, habit tracking, personalized journals, and everyday well-being tools used by thousands of users.
- Designed and developed application components using SwiftUI, Firebase Firestore, and modular state-driven navigation, implementing chatbot flows, journaling modules, onboarding logic, and personalized recommendation systems.
- Developed Sherpa, an empathetic therapy chatbot leveraging OpenAI ChatCompletion API, NLP pipelines, and contextual memory graphs, increasing returning user conversations by 20% through adaptive personalized responses.
- **Tech Stack:** SwiftUI, Firebase Firestore, GCP Cloud Functions, NLP Pipelines, Memory Graphs, REST APIs, Xcode, Git

Graduate Research Assistant - Robotics Software Developer

Jan 2024 – Sep 2024

UIC Human-Centered Engineering Lab

Chicago, IL

- Engineered immersive VR simulations in Unreal Engine (**C++**) using NavigationMesh and physics-based modeling to simulate autonomous delivery robot navigation, pedestrian interactions, and e-scooter dynamics in complex urban environments.
- Partnered with **Honda Research Institute** to identify performance bottlenecks using **Unreal Profiler**. Optimized excessive **CPU usage** and **GUI rendering** overhead by refactoring scene objects, reducing draw calls, resulting in a **27% performance boost**.
- Programmed C++ safety triggers and validated robot behavior using **GSR (Galvanic Skin Response)** data from 50 participant trials.
- Contributed to **HRI code-base architecture** by resolving **55 pull requests**, refactoring simulation modules, documenting APIs, and streamlining version control workflows to enhance reproducibility, collaboration, and system scalability.

Software Development Engineer

Jun 2022 – Aug 2023

LTIMindtree - Link Group - EPlan (Tech Stack – Java, Spring Boot, Rest API, MySQL, Angular, Agile, SDLC)

Hyderabad, India

- Developed an employee stock management **UI** using **Java** and **Spring Boot** featuring participant creation, stock exercises, terminations, vesting, and grant/award management to streamline equity operations, improving process efficiency and data accuracy by 30%.

- Improved the **EPlan** user interface using **Angular**, collaborating with business **stakeholders** to design and refine exercise request, pending request, transaction report, and grant grouping tabs, enhancing navigation flow by 20%.
- Implemented complex **SQL** stored procedures for **performance**-optimized data operations and reporting.
- Established automated test coverage using **JUnit 5** and **Mockito**, enhancing maintainability and reducing production issues by 40%.

Java Spring Boot Microservices Developer

Aug 2021 – May 2022

LTIMindtree - CITI Bank, North America (Tech Stack – Java, Spring boot, Microservices, Rest API, MySQL, Jira)

Hyderabad, India

- Spearheaded the development of an enterprise microservice using **Java** and **Spring Boot** within the Customer Communications Hub (CCH) platform to manage client communications such as account updates, alerts, and KYC document notifications.
- Developed and maintained **RESTful APIs** backed by **MySQL** for the Customer Communications Hub (CCH), optimizing queries to improve data consistency and reduce average response latency by 25% as measured through performance profiling.
- Performed integration testing with **SoapUI**, collaborating with **QA** and **DevOps** teams to improve system reliability and stability by 35%.
- Aligned application implementations with internal security policies and audit requirements to ensure compliance and data protection.

EDUCATION

University of Illinois Chicago

Aug 2023 – May 2025

Master of Science in Computer Science

Chicago, IL

- **Coursework:** Networking and Operating Systems Programming, Distributed Systems, Database Systems, System Design and Scalability

Jawaharlal Nehru Technological University

Aug 2018 – May 2022

Bachelor of Science in Computer Science

Hyderabad, India

- **Coursework:** Data Structures & Algorithms, Advanced Linux Systems, Operating Systems, Object-Oriented Programming (OOP)

PROJECTS

CollabSync – Full-Stack Development and Scalability

Feb 2025 - May 2025

Technologies: JavaScript, React, MongoDB, Node, Express, Firebase, AWS Cloud, Docker

- Designed and implemented an offline-first collaborative editor using React, Node.js, and CRDTs, enabling users to co-author notes and projects without constant internet access and supporting 100 concurrent users with sub-300 ms real-time synchronization.
- Integrated role-based permissions and version history with MongoDB and Firebase, and containerized the system using Docker for deployment on AWS EC2, providing secure collaboration and rollback capabilities during group assignments.

Distributed Multicast MMO Server – System Design and Scalability

Nov 2024 – Dec 2024

Technologies: C++17, Distributed Systems, Boost.Asio, CMake, Linux, TCP/IP, Multithreading

- Achieved world-record level performance in scaling shared-state **distributed server** to **7,000** clients in a single instance.
- Containerized **C++** applications using **Docker** and deployed on **AWS Cloud** utilizing bash scripting and **Docker Swarm**.
- Controlled Virtual IP network of **60+ Linux systems** over UDP/TCP for custom load-testing and server implementation.

MiniOS x86 Kernel – Networking and Operating Systems

Sep 2024 – Oct 2024

Technologies: C, x86 Assembly, Kernel Programming, OS Internals, Paging, Interrupts, Scheduler, Syscalls

- Pioneered a minimal **x86 kernel** in **C/ASM** with bootloader, protected mode entry, and interrupt subsystem (GDT, IDT, PIC, ISR).
- Orchestrated preemptive **round-robin scheduling** with assembly context switching, while developing PS/2 driver, paging, and heap.
- Devised **syscall** interface (int 0x80) with DPL=3, enabling **ring3** programs via TSS/iret to access kernel services (write, yield, sleep).

Stock Watcher – Monthly Quote Retriever

Aug 2024- Oct 2024

Technologies: Bash, awk, GNU date/coreutils, Shell Scripting, Unix/Linux (EWS) environment, Yahoo Finance API

- Automated month-level quote extraction, reducing analyst workload by **95%**, through an epoch-based **Bash** pipeline with leap-year logic.
- Streamlined CSV processing into instantly usable reports under **80ms** by converting dollars to cents and sorting using **awk**.

Empathetic Educational Robot (Misty II) - Research Project

Aug 2023 - Dec 2023

Skills: C#, .NET SDK, Object-Oriented Design, Arduino Backpack, Finite State Machines, Debugging

- Built **C#/.NET SDK** modules with finite state machines for speech modulation and empathetic pauses, improving classroom attentiveness.
- Implemented gestures, LEDs, eye movement, and state-driven Q&A using **OOP design** with Arduino backpack (UART/I²C) integration.
- Designed and executed **NUnit tests** validating speech pacing, gesture synchronization, and edge cases, reducing debugging effort by **25%**.

TECHNICAL SKILLS

Languages and J2EE: Java, Go, Swift, SQL, JDBC, DOM, JSP, JSF, Java Struts, JavaScript

Frameworks and Cloud: Spring Boot, Spring MVC, Eclipse, IntelliJ, VS Code, SoapUI, Angular, GCP, Amazon Web Services (AWS)

Version Control, DevOps and Databases: MySQL, Firebase, Docker, Git, Maven, Gradle, Jenkins, Kubernetes, Mockito